EXP NO:15 Experiment to understand the data preprocessing in Data science

AIM:
To understand the importance of data preprocessing in data science.

PROGRAM:

python

```python
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

# Creating dataset with missing values and categorical data
data = {
    'Country': ['India', 'France', 'Germany', 'France', 'India',
                'Germany', 'India', 'France', 'Germany', 'India'],
    'Age': [44, 27, 30, 38, 40, 35, np.nan, 48, 50, 37],
    'Salary': [72000, 48000, 54000, 61000, np.nan,
               58000, 52000, 79000, 83000, 67000],
    'Purchased': ['No', 'Yes', 'No', 'No', 'Yes',
                  'Yes', 'No', 'Yes', 'No', 'Yes']
}
df = pd.DataFrame(data)
df

# Information about the dataset
df.info()

# Check missing values
df.isnull().sum()

# Handle missing values
df['Age'] = df['Age'].fillna(round(df['Age'].mean()))
df['Salary'] = df['Salary'].fillna(round(df['Salary'].mean()))

# Confirm no missing values remain
df.isnull().sum()

# Encode categorical feature (One-Hot Encoding)
X = pd.get_dummies(df[['Country', 'Age', 'Salary']], columns=['Country'], dtype=int)
X

# Encode the label
y = df['Purchased'].map({'No': 0, 'Yes': 1})
y

# Split into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training set size:", X_train.shape)
print("Testing set size:", X_test.shape)

# Feature scaling
scaler = StandardScaler()
X_train[['Age', 'Salary']] = scaler.fit_transform(X_train[['Age', 'Salary']])
X_test[['Age', 'Salary']] = scaler.transform(X_test[['Age', 'Salary']])

# Preprocessed data
X_train
X_test
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    10 non-null     object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 452.0+ bytes
Training set size: (8, 5)
Testing set size: (2, 5)
```

	Age	Salary	Country_France	Country_Germany	Country_India
8	2.177217	2.300888	0	1	0
1	-2.323996	-1.796813	1	0	0

RESULT:
Thus the program to understand data preprocessing, including handling missing values, encoding, splitting and scaling, was executed successfully.